

Task 4: Generation Parameter Experiments

Prompt Used

"Describe a rainy day in one sentence."

Model Used

gpt-4o-mini

Temperature Comparison Table

Temperature	Run	Output
0.0	1	The sky wept softly as raindrops danced on the pavement, creating a symphony of splashes that mingled with the earthy scent of wet soil and the muted colors of a world cloaked in gray.
0.0	2	The sky wept softly as raindrops danced on the pavement, creating a symphony of splashes that mingled with the earthy scent of wet soil and the muted colors of a world cloaked in gray.
0.0	3	The sky wept softly as raindrops danced on the pavement, creating a symphony of splashes that mingled with the earthy scent of wet soil and the muted whispers of a world cloaked in gray.
0.7	1	Raindrops danced on the pavement, creating a symphony of soft splashes while gray clouds blanketed the sky, wrapping the world in a cozy, melancholic embrace.
0.7	2	A steady drizzle cascades from the gray sky, painting the world in muted hues while the rhythmic patter of raindrops on rooftops creates a soothing symphony of nature.
0.7	3	Raindrops danced on the pavement, creating a symphony of splashes while the gray sky draped the world in a cozy, muted embrace.
1.0	1	The sky wept softly as raindrops danced on the pavement, creating a soothing symphony that blurred the edges of the bustling world around me.
1.0	2	The rhythmic patter of raindrops against the window created a soothing symphony, draping the world in a soft grey haze while the fresh scent of wet earth filled the air.
1.0	3	The rhythmic patter of raindrops against the window created a soothing symphony, as gray clouds draped the world in a soft, misty embrace.

Variance Analysis

1. Temperature 0:

On 0 Temperature all three outputs had the same result. Because low temperature always gives predictable results.

2. Temperature 0.7:

On 0.7 temperature it gave different results but also some words overlap because of little high range temperature we can see the changes in all outputs.

3. Temperature 1:

On 1 temperature it gave more random results but not much change from 0.7. So high temperature gives more random results.

Recommended Temperature by Use Case

1. Support bot

I would choose low temperature because of predictability. Let's assume I want to make a bot for a clinic. So the model will give accurate responses to all patients.

2. Code Generator

In this bot I would choose low temperature for precision and reliability. Low temperature helps to keep syntax same according to requirement. So no logical error will occur.

3. Marketing Copy Tool

In this I would choose high temperature because of randomness. In marketing, we need creative ideas for good marketing. Repetitive lines will make marketing less effective.